

Embedded Systems and IOT

Part B:

Question 1

Reference	Component	Value
V1	DC Source	125V
R1	Resistor	30 Ω
R2	Resistor	50 Ω
R3	Resistor	70 Ω

Question 2

Component	Specification	Function
Arduino Nano	ATmega328P microcontroller board (5V logic)	Acts as the main controller that reads sensor data and processes it
DHT22 Sensor	Digital temperature & humidity sensor (3.3V–5V)	Measures ambient temperature and relative humidity
Resistor	10 Ω	Stabilizes the DATA line for reliable communication (required for bare DHT22)
Jumper Wires	Male-to-male connecting wires	Used to connect Arduino, sensor, and resistor